

# FarSight: Long-Term Disease Prediction Using Unstructured Clinical Nursing Notes

*An overview of the original FarSight paper and how subsequent work extended its contributions.*

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# What Did FarSight Do?

## Primary Focus:

Unstructured **nursing notes** for predicting ICD-9 diagnostic codes.

- Recognized rich, patient-specific info often ignored in modeling.
- Emphasized early-stage detection from clinical observations.

## Techniques Used by FarSight

- Static embeddings extacted through Doc2Vec and variants of NMF-BoWs.

# Our Extensions Beyond FarSight

## Transformer-Based Embeddings

- Shift to **contextual models** like **BioClinicalBERT**
- Captured more **nuanced semantics** from the text

## MoE (Mixture of Experts) Models

- Tackled **data heterogeneity** through domain-specific specialization, enabling tailored processing and integration of diverse data sources.
- Aimed to improve **performance** and **interpretability**

# Results & Model Performance

## NMF-TW with SC's Resilience

- Semantic Coherence helped capture **conceptual relationships**
- **NMF-TW with SC**: Continued strong results (esp. with 100 topics)

## BioClinicalBERT's Contribution

- **BioClinicalBERT**: Promising contextual performance
- Captured **context & long-range dependencies**
- Improved over simpler embedding methods
- Shows possibility of transfer learning in ICD9 prediction tasks.

# Mixture of Experts (MoE) Models

- **MoE Models:** Best for nuanced understanding tasks
- **Heterogeneity:** Experts specialize in data types

**Thank You!**